

The actual conductor covariance: its evaluated outer return and original projection angles

Split-Zero research programme

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The complete outer covariance return is now evaluated through order kq . The original arithmetic kernel satisfies DCR26:

$$\mathcal{R} \log \det H_{K,N}^{\text{ar}} = (16k - 48 - v)q\mathfrak{C}_\partial + \log \frac{\det S_{q-1} \det S_q}{\det S_{2q-1} \det S_{2q}} + o_{A,h}(kq).$$

Each S_N is the specified original source-projection angle matrix, not a freely chosen comparison. The full conductor, invariant image, both physical integration regions, all source minima and four signs remain. The proof pays for every omitted primary row with its full Schur metric bound. The unprojected covariance costs only $O_A(q \log(q+2)) = o(kq)$. The remaining four-angle determinant must still be evaluated to obtain the allocation coefficient; individual complex current signs also remain uncomputed.

These are the complete new proofs CGR1–29, CGS1–22 and DCR1–27. Earlier complete proof files are linked at their precise public editions. The classical Gamma, orthogonal-polynomial and Jensen sources are credited at their use. Finite exact examples check the maps and signs; they do not substitute for the analytic arguments or identify an off-critical zeta zero.

Contents

I	The original covariance and its conductor-pole filtration	4
1	The original invariant covariance and its conductor-pole filtration	4
1.1	1. Every row and every analytic function comes from the original data	4
1.2	2. The exact original Gamma covariance, with its projection retained	5
1.3	3. Finite global bounds needed before eliminating pole directions	6
1.4	4. An exact growing family of conductor-pole constraints	7
1.5	5. The complete projected covariance on those rows has sub-q upper cost	8
1.6	6. The actual allocation loses only $o(kq)$ when these poles are removed	8
1.7	Sources and exact reading coverage	9
 II	 The original strip and its projection-angle loss	 10
2	The original strip has linear elimination depth; its leading native covariance loss is an original projection angle	10
2.1	1. The exact conductor elimination has depth at most $10k$	10
2.2	2. The sparse strip has a smaller interpolation constant before projection	11
2.3	3. The complete unprojected native covariance has only $o(kq)$ determinant cost . .	12
2.4	4. An exact principal-angle formula for the uncomputed native loss	12
2.5	Sources and coverage	13

III	The evaluated outer return and the actual arithmetic kernel	15
3	The original divisor and its complete primary decomposition	15
4	The all-direction estimate for the retained divisor	16
5	Return to the unchanged Gamma metric and observation	18
6	The distant primary components are paid for in the full metric	19
7	The actual graph correction now has one four-angle term	19
8	The receiver in the original arithmetic kernel	20

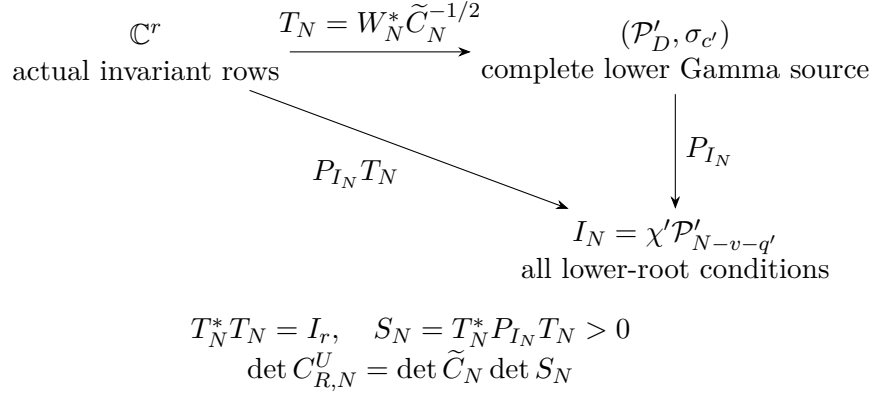


Figure 1: Exact maps CGS13–20. The first arrow is the displayed isometry from the actual invariant covariance, with its full conductor inverse. The second is the unchanged source projection. CGS17 bounds the unprojected determinant by $O_A(q \log(q+2))$; DCR26 preserves the remaining angle determinants at all four cutoffs. No geometric angle is drawn or assigned a value. The complete primary-decomposition diagram appears with DCR5–8.

Part I

The original covariance and its conductor-pole filtration

1 The original invariant covariance and its conductor-pole filtration

21 September 2026. Independent continuation of CG29–32. All coefficient rows below are the actual invariant rows surviving the complete low-jet constraint. No replacement of their metric, their source projection or their original period is made.

This calculation proves two quantitative statements. First, the sum of the positive logarithmic eigenvalues of the original invariant covariance, relative to its specified original coefficient form, is $O_A(q \log(q+2)) = o(kq)$. Second, the remaining graph-covariance changes in CG32 can be restricted to the **actual invariant rows that vanish at the specified conductor poles**, with error $O_A(q\sqrt{k} \log^2(q+2)) = o(kq)$. The removed rows are controlled by proved metric bounds before their number is used. On the retained rows the entire conductor covariance has a quadratic-form upper bound $\exp(o(q))$ in that coefficient form. Its small eigenvalues and its comparison with the outer-ideal covariance remain to be evaluated.

1.1 1. Every row and every analytic function comes from the original data

Retain KF1–55, CG1–32, the complete OCF invariant construction, and

$$N \in \{q-1, q, 2q-1, 2q\}, \quad L = N - q, D = N - v, \quad r = 8k - 32 - v + s_k. \quad (\text{CGR1})$$

Let $Z \in \mathbb{C}^{r \times q}$ contain the actual coefficient rows obtained in CG29–30. It is supported on the OCF strip E_k^{strip} , has full row rank, and each row annihilates the entire degree- $< v$ polynomial space. It is a linear combination of the complete columns $J_R = S_k Z_k$, followed by the exact elimination of $\text{im } B_{\mathcal{L}}$. Its reference coefficient form is

$$R_Z = Z Z^* > 0. \quad (\text{CGR2})$$

This is only a fixed reference used to compare matrices. Every actual Gamma metric below remains unchanged.

For a row $z = (z_\alpha)$ in this original row space define

$$F_z(w) = \sum_{\alpha=1}^q z_\alpha e^{\beta_\alpha w}, \quad G_z(w) = F_z(w)/E_A(w). \quad (\text{CGR3})$$

Since the row kills all degree- $< v$ polynomials, $F_z^{(j)}(0) = 0$ for $j < v$. Thus G_z is holomorphic at zero. The numerator is the original invariant biform restricted to the original exponential curve. OCF17 supplies an invariant lift also for the strip representative: subtracting its conductor part stays inside the invariant image. No arbitrary exponential numerator has been inserted.

For every polynomial p , finite differentiation gives the exact duality

$$[p(\partial_w)F_z(w)]_{w=0} = [(\mathcal{T}_A p)(\partial_w)G_z(w)]_{w=0}. \quad (\text{CGR4})$$

For a single monomial this is Leibniz's formula for $F_z = E_A G_z$; linearity proves the general case. Consequently the complete target observation is

$$\ell_z(\eta) = [\eta(\partial_w)\chi'(\partial_w)G_z(w)]_{w=0}. \quad (\text{CGR5})$$

Indeed KF8–10 give $\mathcal{T}_A \mathcal{R}(\chi' \eta) = \chi' \eta$, and CGR4 gives precisely CG29's coefficient row on every low polynomial η . The extension CGR5 annihilates the **entire** relation:

$$\ell_z(\mathcal{U}_A h) = [(\mathcal{T}_A(\chi h))(\partial_w) G_z]_0 = [(\chi h)(\partial_w) F_z]_0 = 0, \quad (\text{CGR6})$$

because every exponent β_α is a root of the full original χ . Thus it is the actual quotient observation at every cutoff, not an extension chosen for a norm estimate.

1.2 2. The exact original Gamma covariance, with its projection retained

Let $A = A_{D,1}$ be the complete WCF6 matrix from the upper-center Gamma frame $\phi_{v,c}, \dots, \phi_{N,c}$ to the lower-center frame $\phi_{0,c'}, \dots, \phi_{D,c'}$. Its entries and every phase are unchanged. Let $B_F \in \mathbb{C}^{r \times (D+1)}$ have entries

$$(B_F)_{j,n} = [\phi_{v+n,c}(\partial_w) F_{z_j}(w)]_0. \quad (\text{CGR7})$$

In the lower-center orthonormal coefficient space let P_I be the orthogonal projection onto the exact polynomial subspace

$$I_N = \chi' \mathcal{P}'_{L+g} \subset \mathcal{P}'_D. \quad (\text{CGR8})$$

The target observation covariance is exactly

$$\boxed{C_{R,N}^U = B_F A^{-1} P_I (A^{-1})^* B_F^*} \quad (\text{CGR9})$$

To prove it, CGR4 on the source Gamma basis gives the lower functional row $B_F A^{-1}$. Its restriction to I_N is its composition with the orthogonal projection P_I ; the squared Gram of those restricted rows is CGR9. Multiplication by χ' is exactly the isometry from the weighted target source into I_N . CGR6 shows that taking the complete quotient afterward leaves that observation covariance unchanged. The projection P_I , of codimension q' , is present throughout.

Set $R_y = k\sqrt{\delta^2 + \gamma^2}$ and retain the exact finite root-evaluation bound from KF33,

$$V_N = \frac{q e^2}{M_\sigma} (N+1)^{3/2} (2N+1)^{R_y+1}. \quad (\text{CGR10})$$

Then $B_F B_F^* \preceq V_N R_Z$. This follows by writing the rows as the original coefficient rows times the complete root-evaluation matrix; discarding the first v source columns does not increase its norm. The bound itself follows from the generating function on radius $N/(N+1)$, as proved in KF33 and CG23.

With M_D, J_D exactly as in CG23, the WCF19d exterior estimate gives, for each $1 \leq j \leq r$,

$$\prod_{i=1}^j \lambda_i \leq V_N^j M_D^{2(D+1-j)} e^{2J_D}, \quad (\text{CGR11})$$

where $\lambda_1 \geq \dots \geq \lambda_r > 0$ are the eigenvalues of $R_Z^{-1/2} C_{R,N}^U R_Z^{-1/2}$. In fact factor its rectangular Gram matrix as $R_Z^{-1/2} B_F A^{-1} P_I$, apply the exterior norm to each factor, and use that an orthogonal projection contracts every exterior norm. The product of the largest j squared singular values is the displayed left side. Taking the largest partial logarithmic sum proves

$$\boxed{\sum_{i=1}^r \log^+ \lambda_i \leq r \log^+ V_N + 2(D+1) \log M_D + 2J_D = O_A(q \log(q+2)) = o(kq)}. \quad (\text{CGR12})$$

Here $r = O(k)$, $\log V_N = O_A(k \log(q+2))$, and the WCF quantities have their already proved orders. This controls every positively amplified covariance direction jointly. It neither bounds the negative logarithms by the same expression nor assumes the retained projection is an identity.

1.3 3. Finite global bounds needed before eliminating pole directions

These bounds will pay for the actual eliminated directions, including their angles. Let $a_*^{\text{piv}} \neq 0$ be the original leading conductor coefficient selected in OCF5, and put

$$C_A = 1 + A_1/|a_*^{\text{piv}}|, \quad B_{\text{val}} = qM_\sigma \left(\frac{2q + R_y}{2\delta} \right)^{2(q-1)}. \quad (\text{CGR13})$$

Each OCF6 coefficient elimination has coefficient ℓ^1 operator norm at most C_A : it subtracts a scalar coefficient times the full shifted conductor polynomial divided by its actual pivot. There are q' steps. Hence its exact strip reduction satisfies $\|N_k\|_2 \leq \sqrt{q}C_A^{q'}$. For any original strip row z , and any conductor coefficient vector $M_A u$,

$$\|z - M_A u\|_2 \geq \|z\|_2 / \|N_k\|_2. \quad (\text{CGR14})$$

This follows from $N_k M_A = 0$ and $N_k z = z$ in the identical strip positions. It is a quantitative consequence of the actual full conductor elimination, not an assumption about the angle of the invariant image.

The complete source covariance in root-value coordinates is at least $B_{\text{val}}^{-1} I_q$, by the explicit Lagrange interpolation estimate KF34. The lower bound can also be read directly from the inverse norm of the interpolation polynomials: each root separation has modulus at least 2δ , and successive multiplication by $y - \omega$ has Gamma norm at most $2q + R_y$ on the relevant degrees. The sum of their squared norms is exactly bounded by CGR13.

The orthogonal complement of I_N is the span of evaluations at every original lower root. Apply CGR4 to these evaluation rows. WCF21–22 gives their images under A as exactly the complete upper conductor rows. All their first v Gamma coefficients, like those of F_z , vanish. Consequently the distance defining the left side of CGR9 is at least the distance of the original source row to the conductor image divided by $\|A\|$. Using CGR14 and the full source lower bound proves

$$l_{U,N} R_Z \preceq C_{R,N}^U \preceq u_{U,N} R_Z, \quad l_{U,N} = \frac{1}{M_D^2 B_{\text{val}} q C_A^{2q'}}, \quad u_{U,N} = V_N M_D^{2D} e^{2J_D}. \quad (\text{CGR15})$$

For the upper bound use CGR11 at $j = 1$. In particular $\log^+ u_{U,N} + \log^+(1/l_{U,N}) = O_A(q \log(q + 2))$. No lower eigenvalue is asserted to be of order one.

For clarity, the corresponding outer-ideal covariance has finite bounds in the **same** reference form, not a newly chosen observation frame. At $N_0 = q - 1$, CG24 proves $C_{R,N_0}^d = C_{R,N_0}^U$. The outer observation is fixed polynomial remainder followed by A_R , so enlarging its source adds nonnegative rank-one covariance terms. Thus $C_{R,N}^d \succeq l_{U,N} R_Z$ at all later cutoffs.

Here are explicit upper constants. Write $M = L + g$ for the target source degree, $c' = k/2 - 4$, and keep CG6's finite domain. Put

$$\begin{aligned} R_d &= \max\{1, |c'| + q/r_*\}, & t_d &= 2R_d, \\ B_{\text{rem}}(M) &= g^{2g} (1 + R_d)^g \sqrt{M + 1} t_d^M, \\ h_-(M) &= \frac{C_L e^{-\pi(q+1)} (q^2 - R_y^2)^{q'}}{(M + 1)^2 (2M + 1) 64^M (1 + |c'| + q)^{2M}}, \\ h_+^0 &= g M_\sigma (4q + 2 + R_y + |c'|)^{2q}. \end{aligned} \quad (\text{CGR16})$$

Ordinary remainder by d_A/a_A on the coefficient vectors of $1, S', \dots, S'^M$ has norm at most $B_{\text{rem}}(M)$. To prove it, use the exact Cauchy remainder formula on $|S'| = t_d$. Its denominator has modulus at least $(t_d/2)^g$; its monic coefficient sum is at most $(1 + R_d)^g$. The coefficient sum of its divided difference is at most $g t_d^{g-1} (1 + R_d)^g$. Contour length supplies t_d , and a source polynomial has boundary maximum at most $\sqrt{M + 1} t_d^M$ times its coefficient norm. This proves the bound, including repeated roots, since the contour remainder retains all jets.

The complete source Gram H_M in this coefficient frame obeys $H_M \succeq h_-(M)I$. On $q \leq y \leq q+1$, paired lower roots give $|\chi'(c' + iy)|^2 \geq (q^2 - R_y^2)^{q'}$, and CG4's Gamma density bound contributes $C_L e^{-\pi(q+1)}$. Expanding on $[0, 1]$ in shifted Legendre polynomials, as in CG17, bounds the coefficient norm after the substitution $S' = c' + i(q+x)$ by $(M+1)\sqrt{2M+1}8^M(1+|c'|+q)^M$ times the interval L^2 norm. Squaring gives $h_-(M)$. The low Gram H_{g-1} has norm at most h_+^0 : multiply its monomial columns by all q' original root factors and use CG11's upper bound on each, then sum their squared norms. All these products have degree at most q .

At the first cutoff $C_{R,N_0}^U = A_R H_{g-1}^{-1} A_R^*$, hence $A_R A_R^* \preceq h_+^0 u_{U,N_0} R_Z$. Applying the complete remainder bound to the high source covariance now proves

$$l_d R_Z \preceq C_{R,N}^d \preceq u_{d,N} R_Z, \quad l_d = l_{U,N_0}, \quad u_{d,N} = \frac{B_{\text{rem}}(M)^2 h_+^0 u_{U,N_0}}{h_-(M)}. \quad (\text{CGR17})$$

Every logarithmic bound in CGR17 has size $O_A(q \log(q+2))$, since $M \leq q+g = O(q)$, $g = O(k)$, and $R_d = O_A(q)$. These deliberately coarse global constants are used only on the few rows removed next.

1.4 4. An exact growing family of conductor-pole constraints

Retain the full WCF function and the original center difference:

$$w(t) = -i \arctan t, \quad g_A(t) = t^{-v} e^{-4w(t)} E_A(w(t)), \quad g_A(0) = g_0 = (-i)^v \mu_v / v! \neq 0. \quad (\text{CGR18})$$

The powers and logarithm are their analytic branches at $t = 0$, on the unit disk. For the original row z , its upper generating function and the lower generating function of G_z satisfy

$$\begin{aligned} f_z(t) &= t^{-v} (1+t^2)^{-1/4} e^{-cw(t)} F_z(w(t)), \\ H_z(t) &= (1+t^2)^{-1/4} e^{-c'w(t)} G_z(w(t)), \quad f_z(t) = g_A(t) H_z(t). \end{aligned} \quad (\text{CGR19})$$

The exact coefficients of H_z are $[p_n((\partial_w - c')/i) G_z(w)]_0 / n!$. This follows by applying the original Gamma generating function to the linear functional. The factors c, c', i , and e^{-4w} all remain. The apparent singularity of f_z at zero is removable because the first v source coefficients vanish.

Put

$$\eta = k^{-1/2}, \quad \rho = 1 - 2\eta, \quad R_1 = 1 - \eta, \quad R_2 = 1 - \eta/2, \quad (\text{CGR20})$$

which belong to $(0, 1)$ for the original $k \geq 9$. Define

$$M_g = \max \left\{ 1, |g_0|, 2^v A_1 e^{2\pi\gamma} \left(\frac{1+R_2}{1-R_2} \right)^{4\delta} \right\}, \quad J_g = \log(M_g/|g_0|) \geq 0. \quad (\text{CGR21})$$

WCF7–8 bounds $|g_A|$ by M_g on $|t| \leq R_2$. Let its zeros in $|t| < R_1$ be ζ , with their actual multiplicities m_ζ , and let $H_k = \sum m_\zeta$. Jensen's finite disk formula gives

$$H_k \leq \frac{J_g}{\log(R_2/R_1)} \leq \frac{2J_g}{\eta} = O_A(\sqrt{k} \log(k+2)). \quad (\text{CGR22})$$

Indeed every such zero contributes at least $\log(R_2/R_1)$ to the Jensen sum at R_2 , while the boundary mean is at most $\log M_g$. The formula follows by factoring its zeros and applying the mean-value formula to the logarithm of the remaining nonzero analytic factor; this is the same full proof used in OR9. The second inequality follows from $\log(R_2/R_1) \geq (R_2 - R_1)/R_2 \geq \eta/2$.

Define a subspace **inside the original invariant row space** by

$$\mathcal{W}_k^0 = \{z : F_z^{(j)}(w(\zeta)) = 0 \text{ for every } \zeta \text{ above and } 0 \leq j < m_\zeta\}. \quad (\text{CGR23})$$

Its exact codimension h_k is the rank of the displayed finite matrix of actual values $\beta_\alpha^j e^{\beta_\alpha w(\zeta)}$ after multiplication by Z ; in particular $h_k \leq H_k$. Every derivative and point is specified by the original conductor and invariant rows. Since $w'(t) = -i/(1+t^2) \neq 0$, these constraints say exactly that the numerator f_z cancels the indicated zeros of g_A with their full multiplicities. There is no unspecified generic rank.

1.5 5. The complete projected covariance on those rows has sub-q upper cost

Let $B(t)$ be the finite disk Blaschke product of the zeros in CGR22, with factors $R_1(t-\zeta)/(R_1^2-\bar{\zeta}t)$, repeated with their multiplicities. Its boundary modulus on $|t| = R_1$ is one and $|B(0)| \leq 1$. Both f_z/B and g_A/B are holomorphic there for $z \in \mathcal{W}_k^0$, and the second has no interior zero. The maximum principle gives $\max |f_z/B| \leq \max_{|t|=R_1} |f_z(t)|$. Also $u(t) = \log(M_g/|g_A(t)/B(t)|)$ is nonnegative harmonic in that disk, with $u(0) \leq J_g$. Its Poisson integral gives

$$u(t) \leq \frac{R_1 + \rho}{R_1 - \rho} u(0) \leq \frac{2J_g}{\eta} \quad (|t| \leq \rho). \quad (\text{CGR24})$$

For the inequality, the Poisson kernel on any intermediate circle of radius $R < R_1$ is at most $(R + |t|)/(R - |t|)$; its average of u is $u(0)$. Let R tend to R_1 . Boundary zeros of g_A , if any, do not affect this interior argument or the maximum principle.

The numerator bound retaining its original coefficient norm is

$$\max_{|t|=R_1} |f_z(t)| \leq F_k \|z\|_2, \quad F_k = R_1^{-v} (1 - R_1^2)^{-1/4} \sqrt{q} \exp\{R_y \operatorname{arctanh} R_1\}. \quad (\text{CGR25})$$

In fact the original center factor cancels e^{cw} in each original exponential; the remaining exponent has modulus at most $R_y|w|$, and $|w(t)| \leq \operatorname{arctanh} R_1$. Cauchy-Schwarz on all q unchanged coefficients proves CGR25. Using $M_g \geq 1$, CGR24–25 therefore bound the full lower generating function by $\max_{|t| \leq \rho} |H_z(t)| \leq F_k e^{2J_g/\eta} \|z\|_2$. Its coefficient of degree n has modulus at most this quantity times ρ^{-n} . The normalized Gamma functional coefficient multiplies that coefficient by $\rho_n/\sqrt{M_\sigma}$, where $\rho_n^2 = n!/(1/2)_n \leq 2n+1$. Summing every source coefficient through D , and then retaining the original orthogonal projection, proves

$$\boxed{C_{R,N}^U|_{\mathcal{W}_k^0} \preceq e^{\varepsilon_{k,N}} R_Z|_{\mathcal{W}_k^0}}, \quad \varepsilon_{k,N} = \log \frac{(D+1)(2D+1)}{M_\sigma} + 2 \log F_k + \frac{4J_g}{\eta} + 2D \log(1/\rho). \quad (\text{CGR26})$$

An upper bound using $\max(0, \varepsilon_{k,N})$ is available if a nonnegative allowance is desired. Explicitly

$$\varepsilon_{k,N} = O_A(k^{3/2} + k \log(k+2) + \sqrt{k} \log(k+2)) = o(q). \quad (\text{CGR27})$$

Here $D \leq 2q$, $\log(1/\rho) = O(k^{-1/2})$, $\operatorname{arctanh} R_1 = O(\log k)$, and $J_g = O_A(\log k)$. The proof never replaces the projected norm by an equality with the full norm; the full norm is used only for its valid upper bound.

1.6 6. The actual allocation loses only $o(kq)$ when these poles are removed

Choose a basis orthonormal for the fixed coefficient form R_Z , whose first $r - h_k$ rows span the exact subspace CGR23. Use that same basis at both high cutoffs and for both U, d . In its block decomposition the determinant equals the determinant of the first restriction times the determinant of its attained Schur complement. Denote the covariance change on the first restriction by $\Delta_{R,N}^0$.

Congruence and attained minimization transport CGR15 and CGR17 to those Schur complements: every eigenvalue for U lies in $[l_{U,N}, u_{U,N}]$, and every one for d lies in $[l_d, u_{d,N}]$. Their rank is exactly h_k . Therefore

$$\boxed{|\Delta_{R,N} - \Delta_{R,N}^0| \leq h_k \max \left\{ \log^+ \frac{u_{U,N}}{l_d}, \log^+ \frac{u_{d,N}}{l_{U,N}} \right\} = O_A(q\sqrt{k} \log^2(q+2)) = o(kq)}. \quad (\text{CGR28})$$

This is where the number of pole directions is used. Their complete metric costs were bounded first, so no unproved angle estimate or deletion by rank is concealed in CGR28. The coefficient basis is only an exact fixed change of frame and its determinant cancels in each covariance ratio.

Combining with the already evaluated full-volume error CG32 gives

$$\boxed{\mathcal{G}_K = -\Delta_{R,2q-1}^0 - \Delta_{R,2q}^0 + o_A(kq).} \quad (\text{CGR29})$$

For a finite bound, add CG32's displayed finite right side to the two finite right sides of CGR28. Thus the remaining leading allocation is now localized to an explicitly defined, original invariant row space with every selected conductor pole canceled. It still has dimension $r - O_A(\sqrt{k} \log(k+2))$. CGR26–27 control its upper covariance quadratic form. The lower spectrum and the signed comparison with $C_{R,N}^d$ on this space are not evaluated here; an upper bound alone cannot assign their determinant ratio or an RH conclusion.

1.7 Sources and exact reading coverage

OCF1–26, *An exact conductor frame and a fixed invariant-generator recursion*, was read completely for this continuation. CGR3 and CGR14 use its exact strip representatives, conductor columns and invariant certificates. The complete pinned proof file is byte-verified in the root public-source receipt.

KF1–55 and CG1–32 were read completely. This continuation retains their actual A_R , full low constraint, frame and source. Their complete source must accompany publication before a live link is claimed.

WCF4–13, WCF19a–f and WCF20–22 are used from the pinned complete proof file, whose byte identity was checked in the CG source. The cited portions were read directly. The Gamma generating function is NIST DLMF 18.23.7, with recurrence 18.22.8, by T. H. Koornwinder, R. Wong, R. Koekoek, R. F. Swarttouw and W. P. Reinhardt, version 1.2.8. Its original TeX encodings are retained in `graph_primary_sources/`. The generating function is applied explicitly in CGR19 with both original centers and their exact phase. Jensen's zero count is the classical formula of J. L. W. V. Jensen, *Sur un nouvel et important théorème de la théorie des fonctions* (1899), DOI 10.1007/BF02417878; the finite proof used here is included in OR9 and recalled in CGR22. The disk-product and harmonic inequalities needed in CGR24 are proved above rather than supplied as missing analytic assumptions.

The independent checker `check_invariant_covariance.py` passed 96 exact tests of CGR3–9, with $v=0$ and $v=1$ finite-shift fixtures, both retained Gamma centers, the physical complex phase, the complete relation minimum and the original source projection. Deleting that projection or changing the phase fails the respective negative control. These finite checks do not replace the analytic proofs of CGR10–29.

The exact earlier proof files are KF1–55, CG1–32, and OR1–25. CGR and CGS are reproduced completely in this edition.

Part II

The original strip and its projection-angle loss

2 The original strip has linear elimination depth; its leading native covariance loss is an original projection angle

21 September 2026. Independent continuation of CGR1–29. The full original conductor, invariant image, Gamma source and quotient minimum are retained. The results below concern the actual row space of CG29–30 at all four original cutoffs. They quantify the source contribution and locate the remaining leading covariance in an exact projection. They do not assign the remaining projection determinant a value.

2.1 1. The exact conductor elimination has depth at most $10k$

Use the entire OCF4 conductor

$$A = \sum_{r,s=0}^8 a_{rs} e_{rs}^{(8)}, \quad (r_0, s_0) = \max_{\text{lex}} \{(r, s) : a_{rs} \neq 0\}, \quad a_* = a_{r_0 s_0}, \quad A_1 = \sum_{r,s} |a_{rs}|. \quad (\text{CGS1})$$

The subscripts r_0, s_0 here denote the actual leading monomial, not the earlier residual-rank symbol. Set

$$B_A^{\text{step}} = \frac{A_1 - |a_*|}{|a_*|}, \quad C_A^{\text{path}} = 1 + B_A^{\text{step}} = \frac{A_1}{|a_*|} \geq 1. \quad (\text{CGS2})$$

Give the original coefficient position $(a, b) \in \{0, \dots, k\}^2$ the integer height $h(a, b) = 9a + b$. OCF6 replaces a pivot coefficient at $(r_0 + i, s_0 + j)$ by the entire sum

$$- \sum_{(r,s) \neq (r_0, s_0)} \frac{a_{rs}}{a_*} e_{r+i, s+j}^{(k)}. \quad (\text{CGS3})$$

Every nonzero term has strictly smaller height. If $r = r_0$, its decrease is $s_0 - s \geq 1$. If $r < r_0$, the decrease is $9(r_0 - r) + (s_0 - s) \geq 9 - 8 = 1$. Every destination remains in the original coefficient box, since $0 \leq i, j \leq k - 8$ and $0 \leq r, s \leq 8$. Thus a path of successively replaced coefficients has length at most $10k$; coefficients are never removed from the original box to obtain this bound.

For completeness, expand the reduction of one basis column as the sum over all its substitution paths, stopping each path when it reaches the actual strip. At a substituted node the sum of absolute edge weights is B_A^{step} . The sum of absolute weights of all terminal paths of a specified length j is consequently at most $(B_A^{\text{step}})^j$, by induction on the successive levels. The triangle inequality therefore bounds the total output column norm by $\sum_{j=0}^{10k} (B_A^{\text{step}})^j$, which is at most $(1 + B_A^{\text{step}})^{10k}$. The latter assertion follows by expanding the binomial and using $\binom{10k}{j} \geq 1$. When $B_A^{\text{step}} = 0$, the same expression is one. The exact strip map therefore obeys

$$\|N_k\|_{1 \rightarrow 1} \leq (C_A^{\text{path}})^{10k}, \quad \|N_k\|_{2 \rightarrow 2} \leq \sqrt{q} (C_A^{\text{path}})^{10k}. \quad (\text{CGS4})$$

This computes a quantitative bound for the existing OCF elimination; the order and coefficients of that elimination have not been changed. The second inequality uses $\|u\|_2 \leq \|u\|_1$ on the output and $\|f\|_1 \leq \sqrt{q} \|f\|_2$ on its original input.

In the identical strip positions, $N_k z = z$ and $N_k M_A = 0$. Thus the complete conductor image has the following proved angle from every original strip coefficient vector:

$$\text{dist}_{\ell^2}(z, \text{im } M_A) \geq \frac{\|z\|_2}{\sqrt{q} (C_A^{\text{path}})^{10k}}. \quad (\text{CGS5})$$

Apply N_k to $z - M_A u$, use CGS4, and take the infimum over all original conductor coefficients u to prove the formula. Consequently CGR15 remains valid with the path-based lower constant

$$l_{U,N}^{\text{path}} = \left[M_D^2 B_{\text{val}} q (C_A^{\text{path}})^{20k} \right]^{-1}. \quad (\text{CGS6})$$

The B_{val} in this statement is still the complete q -root interpolation constant of CGR13. The original source projection is still present; CGS5 does not erase that factor. One may take the larger of CGS6 and the prior lower constant at each finite cutoff. The new path bound improves the growth exponent as k increases; it need not be the smaller numerical bound for the first few allowed values of k .

2.2 2. The sparse strip has a smaller interpolation constant before projection

There are exactly

$$n_E = |E_k^{\text{strip}}| = q - q' = 16k - 48 \quad (\text{CGS7})$$

original roots in the strip. Their physical coordinates are

$$y_{ab} = (\beta_{ab} - c)/i = (2b - k)\gamma - i(2a - k)\delta, \quad (a, b) \in E_k^{\text{strip}}, \quad c = k/2. \quad (\text{CGS8})$$

Every one has modulus at most $R_y = k\sqrt{\delta^2 + \gamma^2}$. Every pair has separation at least 2δ , because $0 < \delta < 1/2$, $\gamma > 2$, and distinct pairs differ in at least one coordinate. In the original Gamma norm define its Lagrange polynomials by

$$L_\alpha(y) = \prod_{\beta \in E_k^{\text{strip}}, \beta \neq \alpha} \frac{y - y_\beta}{y_\alpha - y_\beta}. \quad (\text{CGS9})$$

These interpolate precisely the strip roots. No lower-lattice root is inserted in this interpolation and none is dropped from the later source projection.

Successive application of CG11 to every displayed factor gives

$$\|L_\alpha\|_\sigma^2 \leq M_\sigma \left(\frac{2n_E + R_y}{2\delta} \right)^{2(n_E-1)}, \quad B_E = n_E M_\sigma \left(\frac{2n_E + R_y}{2\delta} \right)^{2(n_E-1)}. \quad (\text{CGS10})$$

Indeed the degree before any multiplication is at most $n_E - 2$, so CG11 bounds that multiplication norm by $2n_E + R_y$; each denominator has modulus at least 2δ , and the norm of the constant polynomial is exactly $\sqrt{M_\sigma}$.

Let $E_{E,N}$ evaluate the complete upper Gamma frame of degrees $0, \dots, N$ at these actual roots. Since $n_E - 1 \leq q - 1 \leq N$, the polynomials CGS9 form a right inverse of this evaluation map at every original cutoff. Its squared operator norm is at most the sum of the squared column norms, namely B_E . For a row z , applying that right inverse to $zE_{E,N}$ gives

$$\|zE_{E,N}\|_2^2 \geq B_E^{-1} \|z\|_2^2. \quad (\text{CGS11})$$

This proves the lower bound without a Vandermonde determinant estimate.

The actual rows in Z are supported on this strip and annihilate all polynomials of degree below v . Their first v Gamma functional coefficients consequently vanish exactly. Thus removing those zero columns leaves the identical row norm, and CGR7 satisfies

$$\boxed{B_E^{-1} R_Z \preceq B_F B_F^* \preceq V_N R_Z.} \quad (\text{CGS12})$$

The right inequality is CGR10. The left inequality uses exactly the unchanged sparse rows of the invariant image. It is not a statement about the covariance after projecting away the lower root evaluations.

2.3 3. The complete unprojected native covariance has only $o(kq)$ determinant cost

Retain $A = A_{D,1}$, $D = N - v$, and define

$$W_N = B_F A^{-1}, \quad \tilde{C}_{R,N} = W_N W_N^*, \quad C_{R,N}^U = W_N P_I W_N^*. \quad (\text{CGS13})$$

Every matrix is in the original lower-center Gamma frame; CGR9 proves the last equality with the original orthogonal projection P_I onto $\chi' \mathcal{P}'_{L+g}$. The forward WCF bound $\|A\| \leq M_D$ and CGS12 prove

$$\tilde{C}_{R,N} \succeq (M_D^2 B_E)^{-1} R_Z. \quad (\text{CGS14})$$

For each $j \leq r$, the same exterior argument as CGR11, now without a projection, gives

$$\prod_{i=1}^j \tilde{\lambda}_i \leq V_N^j M_D^{2(D+1-j)} e^{2J_D}, \quad (\text{CGS15})$$

where the ordered eigenvalues are those relative to R_Z . All factors come from the original operator and the exact strip rows.

Set

$$\begin{aligned} K_{-,N} &= r \log^+(M_D^2 B_E), \\ K_{+,N} &= r \log^+ V_N + 2(D+1) \log M_D + 2J_D, \\ K_N &= \max\{K_{-,N}, K_{+,N}\}. \end{aligned} \quad (\text{CGS16})$$

Taking the determinant in CGS14 bounds its logarithm below by $-K_{-,N}$. Taking the largest partial sum of the ordered logarithms in CGS15 bounds its positive logarithmic part by $K_{+,N}$, hence its whole logarithm above by that number. We have therefore evaluated the size of the complete source contribution:

$$\left| \log \frac{\det \tilde{C}_{R,N}}{\det R_Z} \right| \leq K_N = O_A(q \log(q+2)) = o(kq). \quad (\text{CGS17})$$

To check the stated order, $r = O(k)$, $n_E = O(k)$, $R_y = O_A(k)$, $\log B_E = O_A(k \log(k+2))$, $\log V_N = O_A(k \log(q+2))$, and $\log M_D = O_A(\log(q+2))$, $J_D = O_A(q)$ are the proved bounds of the cited formulas. The endpoints satisfy $D \leq 2q$. Thus both finite quantities in CGS16 are $O_A(q \log(q+2))$; since $q = (k+1)^2$, division by kq tends to zero.

2.4 4. An exact principal-angle formula for the uncomputed native loss

The following maps give the geometry of the remaining term, with their domains and codomains retained:

$$\mathbb{C}^r \xrightarrow{T_N = W_N^* \tilde{C}_{R,N}^{-1/2}} \mathcal{P}'_D \xrightarrow{P_I} I_N = \chi' \mathcal{P}'_{L+g} \subset \mathcal{P}'_D. \quad (\text{CGS18})$$

The middle space has the original center- c' Gamma inner product and its orthonormal coefficient frame. The first map is an isometry: $T_N^* T_N = I_r$ follows directly from CGS13. The second map is the original orthogonal projection and uses every lower-root condition. Consequently

$$S_N = T_N^* P_I T_N = \tilde{C}_{R,N}^{-1/2} C_{R,N}^U \tilde{C}_{R,N}^{-1/2} \quad \text{satisfies} \quad 0 < S_N \preceq I_r. \quad (\text{CGS19})$$

Strict positivity follows because the original quotient observation has full row rank: if a combination of its rows has zero norm on I_N , it vanishes on the low target polynomials too, contrary to the full row rank of A_R . The eigenvalues of S_N are the squares of the singular values of $P_I T_N$, or equivalently the squared cosines of the principal angles between $\text{im } T_N$ and I_N . This equivalence follows from the defining Gram $(P_I T_N)^* (P_I T_N) = S_N$; it introduces no assumed angle bound.

Taking determinants in CGS19 and applying CGS17 proves the finite quantitative receiver

$$\left| \log \frac{\det C_{R,N}^U}{\det R_Z} - \log \det S_N \right| \leq K_N = o_A(kq). \quad (\text{CGS20})$$

Thus the entire leading native covariance loss is the sum of logarithms of these original squared projection angles. Every contribution from the unprojected source, including its actual invariant coefficients and the full conductor inverse, has been bounded at the required scale.

For the two high endpoints, put $D_{d,N} = \log(\det C_{R,N}^d / \det R_Z)$, with the unchanged outer-ideal covariance of CG31. The earlier complete graph calculation then gives

$$\left| \mathcal{G}_K - \sum_{N \in \{2q-1, 2q\}} (D_{d,N} - \log \det S_N) \right| \leq B_{\text{CG32}} + K_{2q-1} + K_{2q} = o_A(kq), \quad (\text{CGS21})$$

where B_{CG32} is exactly the finite right side of CG32. To verify the sign, CG31 defines $\Delta_{R,N} = \log \det C_{R,N}^U - \log \det C_{R,N}^d$, and CG32 gives $\mathcal{G}_K = -\sum_N \Delta_{R,N}$ up to its displayed error. Insert CGS20 into this identity.

Every statement CGS12–20 also applies to the specified pole-canceling row subspace $\mathcal{W}_k^0$ of CGR23, using the same restricted coefficient form and its actual row dimension. On that subspace CGR26 also bounds the unprojected covariance, because its proof bounds the full Gamma functional before using the projection. Hence

$$(M_D^2 B_E)^{-1} R_Z|_{\mathcal{W}_k^0} \preceq \tilde{C}_{R,N}|_{\mathcal{W}_k^0} \preceq e^{\varepsilon_{k,N}} R_Z|_{\mathcal{W}_k^0}. \quad (\text{CGS22})$$

Both logarithmic quadratic-form allowances are $o(q)$, with all constants given in CGS10 and CGR26. Combining the subspace version of CGS21 with CGR28 retains its additional, already proved $O_A(q\sqrt{k}\log^2(q+2))$ error. The unresolved leading quantity is now precisely the relative outer covariance versus these original projection angles. No value or sign for that quantity is asserted by the present bound.

2.5 Sources and coverage

OCF1–26, *An exact conductor frame and a fixed invariant-generator recursion*, was read completely. CGS1–5 use OCF4–8 with the identical coefficients, pivot and original strip. The complete public proof file has a byte-verification receipt in the parent source ledger.

KF1–55, CG1–32 and CGR1–29 were read completely. The source files are `ACTUAL_KERNEL_FILTRATION_AND_GRAPH.md`, `GRAPH_METRIC_DERIVATION.md` and `INVARIANT_COVARIANCE_POLE_FILTRATION.md`, respectively. Their complete sources must accompany public use when a pinned proof-file link has not yet been verified. Repeated equation prefixes in other documents do not identify these sources.

The WCF forward and exterior bounds are WCF4–13 and WCF19a–f in the complete pinned proof file. The cited portions were read directly, and the public source was checked byte for byte against the local source used here. CG11 derives its Gamma factor estimate from the Meixner–Pollaczek recurrence and generating function in NIST DLMF 18.22.8 and 18.23.7, by T. H. Koornwinder, R. Wong, R. Koekoek, R. F. Swarttouw and W. P. Reinhardt. The original NIST TeX encodings are retained and read in `graph_primary_sources/`. The path-depth proof, all interpolation inequalities and the exact projection-angle map used here are proved in this file.

The independent `check_invariant_strip.py` passed 282 exact checks. They include all 81 possible leading-monomial height cases; the complete original decreasing-lex elimination at eight auxiliary conductor fixtures; the sparse Gamma interpolation estimate; and the exact retained projection-angle determinant. A negative control verifies that the projection determinant cannot be set equal to one. The finite fixtures support the algebraic identities and do not evaluate the remaining original growing-degree determinant.

The exact earlier proof files are KF1–55, CG1–32, and OR1–25. CGR and CGS are reproduced completely in this edition.

Part III

The evaluated outer return and the actual arithmetic kernel

The earlier outer-quotient estimate OOQ1–40 applies to a divisor whose roots are all small relative to the source scale. The complete conductor divisor has a few additional distant roots. We retain their primary components, bound their complete metric cost, and prove the outer covariance return for the actual invariant observation. This evaluates one of the two terms left in CGS21. The remaining term is a specified four-cutoff determinant of the original projection angles.

All assertions use the fixed original conductor and period on the simple-quartet domain of KF1–10 and CG1–6. The equilibrium input and source comparison used below are the proved statements EIQ1–33 and OOQ3–11, OOQ30–31 in the cited complete source [1]; their previously recorded mathematical status is unchanged.

3 The original divisor and its complete primary decomposition

Retain

$$q = (k + 1)^2, \quad q' = (k - 7)^2, \quad v = \text{ord}_0 E_A \leq 80, \quad g = q - q' - v, \quad c' = k/2 - 4.$$

The conductor polynomial is exactly

$$d_A(S') = \frac{\mathcal{T}_A \chi(S')}{\chi'(S')}, \quad a_A = [S'^g]d_A = \mu_v \binom{q}{v} \neq 0.$$

Here χ' names the lower root polynomial. The precise change to the physical coordinate is $S' = c' + iy$, and put

$$p_A(y) = \frac{d_A(c' + iy)}{a_A i^g}, \quad E_d = \mathbb{C}[S']/(d_A) \xrightarrow{f \mapsto f(c' + iy)} \mathbb{C}[y]/(p_A). \quad (\text{DCR1})$$

This is an algebra isomorphism; multiplication of a defining polynomial by its displayed nonzero scalar does not change the ideal. No source norm is divided by that scalar. At cutoff N , the source degree and norm are

$$M = N - q + g, \quad \|f\|_H^2 = \int_{\mathbb{R}} |f(c' + iy)|^2 |\chi'(c' + iy)|^2 d\sigma(y), \quad d\sigma(y) = \frac{|\Gamma(1/4 + iy/2)|^2}{2\pi} dy. \quad (\text{DCR2})$$

Thus $M = g - 1, g, q + g - 1, q + g$ at the four original cutoffs. Let Q_N^d be its attained remainder metric in E_d , and keep the actual fixed onto map $A_R : E_d \rightarrow \mathbb{C}^r$ of CG29–30, where $r = 8k - 32 - v + s_k$. Its covariance is $C_{R,N}^d = A_R(Q_N^d)^{-1} A_R^*$. All references to this map include the complete invariant image. For its literal coordinate transport let $T_{c',i}$ be the g -by- g matrix with entries $(T_{c',i})_{aj} = \binom{j}{a} (c')^{j-a} i^a$ for $a \leq j$, and zero otherwise. It sends the coefficients of $f(S')$ to those of $f(c' + iy)$; its inverse sends y^j to $i^{-j} (S' - c')^j$. Hence the same actual observation has matrix $A_R^y = A_R T_{c',i}^{-1}$ in physical remainder coordinates. In every CRT expression below, the abstract maps π_b, ι_b, ι_f include this displayed isomorphism to or from E_d . Equivalently their coefficient formulas use A_R^y in place of the original S' -coordinate matrix. This specifies the exact map without identifying the two Grams.

Let $R_b = q^{7/8}/8$. Factor the entire monic physical polynomial as

$$p_A = p_b p_f, \quad p_b = \prod_{|\zeta| < R_b} (y - \zeta), \quad p_f = \prod_{|\zeta| \geq R_b} (y - \zeta), \quad a = \deg p_b = g - b, \quad b = \deg p_f. \quad (\text{DCR3})$$

Every factor is included with its full multiplicity. Roots on the circle belong to p_f . These factors are coprime. Under the explicit OR12 guards, OR13 and OR15 give

$$0 \leq b \leq b_{\max} := \frac{\log^+(A_1/a_*) + 32Bq^{1/8}}{\log 2} = O_A(q^{1/8}), \quad |\zeta| < q/r_* \text{ for every root.} \quad (\text{DCR4})$$

The quantities A_1, a_*, B, r_* are precisely OR4, OR7 and OR14. In particular $a > 0$ eventually; this is one of the finite guards below.

The original primary decomposition is the Chinese-remainder map

$$\mathbb{C}[y]/(p_A) \longrightarrow E_b \oplus E_f, \quad E_b = \mathbb{C}[y]/(p_b), \quad E_f = \mathbb{C}[y]/(p_f). \quad (\text{DCR5})$$

To specify its inverse, choose the unique remainder u_b with $u_b p_f = 1 \pmod{p_b}$. The section $\iota_b[f] = [u_b p_f f]_{p_A}$ has first component f and second component zero; interchange the factors for ι_f . Bezout's polynomial division identity proves these statements, also at repeated roots. The kernel of the original remainder map $\pi_b : E_d \rightarrow E_b$ is the b -dimensional space $\iota_f E_f$.

Among the actual observation rows define

$$\mathcal{V}^0 = \{\lambda \in (\mathbb{C}^r)^* : \lambda A_R \iota_f = 0\}, \quad h = r - \dim \mathcal{V}^0 = \text{rank}(A_R \iota_f) \leq b. \quad (\text{DCR6})$$

Choose a row basis B_0 of this exact subspace once for all cutoffs. There is an onto map $A_b : E_b \rightarrow \mathbb{C}^{r-h}$ such that

$$B_0 A_R = A_b \pi_b, \quad A_b = B_0 A_R \iota_b. \quad (\text{DCR7})$$

Indeed DCR5 gives the factorization, and independent rows of $B_0 A_R$ prove surjectivity. The source observation is consequently the literal map $f \mapsto A_b \text{rem}_{p_b} f(c' + iy)$. Taking the minimum over these identical fibres proves

$$B_0 C_{R,N}^d B_0^* = A_b (Q_{b,M})^{-1} A_b^*, \quad (\text{DCR8})$$

where $Q_{b,M}$ is the quotient metric by p_b with the *entire* original weight in DCR2. This equality retains the distant roots through the exact choice DCR6; their remaining rows are estimated in Section 4.

4 The all-direction estimate for the retained divisor

We give explicit bounds rather than assume that OOQ's original divisor can be substituted. Use exactly its equilibrium constants $a_*, C_{\text{BM}}, C_W, C_{\text{quant}}, C_0, C_{\text{tail}}, T_*$ from OOQ4–6; only in this section the symbol a_* is the equilibrium endpoint minimum of OOQ4. To avoid confusing it with OR7 in DCR4, write it as a_{eq} below. Put

$$n = q/2, \quad \rho = n^{-1/16}, \quad r_b = R_b/n, \quad \vartheta = \pi/2, \quad Q = q', \quad \beta_{\pm} = \vartheta \pm 1/q. \quad (\text{DCR9})$$

Retain the original OSP domain used in OOQ31 and the OR12/CG6 domains. Impose also $a \geq 1$, $n \geq 2g + 1$, $n \geq 2/b_*$, $r_b \leq \rho/2$, $\rho \leq \min(1/2, \sqrt{a_{\text{eq}}/2})$, $k \geq 257$, and that the three parameter pairs

$$((Q - \tfrac{1}{2})/n, \beta), \quad ((Q - \tfrac{1}{2})/m_e, n\beta/m_e), \quad ((Q + \tfrac{1}{2})/m_o, n\beta/m_o)$$

lie in the explicit rectangle $[3/2, 5/2] \times [\pi/4, \pi]$, for both rates and both high source degrees, where $m_e = \lfloor M/2 \rfloor$, $m_o = \lfloor (M-1)/2 \rfloor$. These are finite inequalities on the original data. They all hold eventually: $g = O(k)$, $b = O_A(q^{1/8})$, $r_b = q^{-1/8}/4$, and $q' = q - O(k)$. No conclusion at a cutoff outside these guards is inferred.

Let $\widehat{p}_b(z) = n^{-a} p_b(nz)$. For an analytic function F on the closed radius- ρ disk the exact remainder is

$$\mathcal{R}_b F(z) = \frac{1}{2\pi i} \int_{|\zeta|=\rho} F(\zeta) \frac{\widehat{p}_b(\zeta) - \widehat{p}_b(z)}{(\zeta - z)\widehat{p}_b(\zeta)} d\zeta, \quad \|\mathcal{R}_b F\|_1 \leq C_b \max_{|\zeta|=\rho} |F(\zeta)|, \quad C_b = \rho a (1 + r_b)^a (2/\rho)^a. \quad (\text{DCR10})$$

For the identity, subtract the integral from Cauchy's formula for $F(z)$; their difference is $\widehat{p}_b(z)$ times an analytic function. Thus the remainder matches every derivative through the multiplicity of each root. Its degree is below a , so the polynomial identity holds including collisions. For the bound, the denominator has modulus at least $(\rho/2)^a$; the coefficient sum of the divided difference is at most $a(1+r_b)^a$; the contour contributes ρ . This is the complete repeated-root version of OOQ14–15.

First use the exact exponential norm $\int |f(y)|^2 |y|^{2Q} e^{-\beta|y|} dy$. Put $\alpha_0 = (Q - \frac{1}{2})/n$, $W = W(\alpha_0, \beta)$, and

$$c_{\text{high}}(Q, \beta) = (2Q + 1) \log n - 2nW.$$

OOQ7–8 is an evaluation bound for *any* polynomial, independent of a divisor. Write the unchanged scaling $f(nz) = f_e(z^2) + zf_o(z^2)$. Its norm is exactly

$$n^{2Q+1} \left[\int_0^\infty |f_e(x)|^2 x^{Q-1/2} e^{-n\beta\sqrt{x}} dx + \int_0^\infty |f_o(x)|^2 x^{Q+1/2} e^{-n\beta\sqrt{x}} dx \right].$$

The two halves of the original real line supply this identity. Since $m_e, m_o = n + O(g)$, the OOQ6 derivative bound gives $|m_e W_e - nW|, |m_o W_o - nW| \leq C_W(g+2)$. Apply OOQ8 on $x = \zeta^2$ and then DCR10. Every lift of a scaled remainder vector t therefore satisfies

$$\|t\|_2 \leq C_b \sqrt{2} C_{\text{BM}}(n+g)^2 e^{nW+C_W(g+2)+2(n+g)\rho^2/a_{\text{eq}}} n^{-(2Q+1)/2} \|f\|_{\text{exp}}. \quad (\text{DCR11})$$

For the matching upper bound use the *actual* OOQ9 quantile polynomial $r_n(x)$, and set

$$a_t(z) = \mathcal{R}_b\left(\frac{t(z)}{r_n(z^2)}\right), \quad f_t(nz) = r_n(z^2) a_t(z). \quad (\text{DCR12})$$

Its degree is at most $q + a - 1 \leq q + g - 1$. The denominator is nonzero on the disk. DCR10 proves equality of every required root derivative, so it is a genuine lift. OOQ10–11 and DCR10 give

$$\|a_t\|_1 \leq C_b \sqrt{a} e^{2n\rho^2/a_{\text{eq}}} \|t\|_2, \\ \|f_t\|_{\text{exp}}^2 \leq e^{c_{\text{high}}} \|a_t\|_1^2 \exp\{2C_{\text{quant}}\sqrt{n} + 2C_0 + 2C_{\text{tail}}(a-1)\} T_*.$$

To verify the latter estimate, use both real branches $|a_t(\pm\sqrt{x})| \leq \|a_t\|_1(1+\sqrt{x})^{a-1}$. The full OOQ Euler identity gives the integrand factor $e^{-nD(x)}$. Its remaining multiplier is at most $e^{2C_{\text{tail}}(a-1)} e^{2(a-1)D(x)}$. The guard $n - 2(a-1) \geq 1$ leaves the integrable majorant $e^{-D(x)}$ with integral at most T_* . Thus both integration regions and all source powers remain.

Define

$$E_H^- = 2 \log(C_b \sqrt{2} C_{\text{BM}}(n+g)^2) + 2C_W(g+2) + 4(n+g)\rho^2/a_{\text{eq}}, \\ E_H^+ = 2 \log C_b + \log a + 4n\rho^2/a_{\text{eq}} + 2C_{\text{quant}}\sqrt{n} + 2C_0 + 2C_{\text{tail}}(a-1) + \log T_*, \quad (\text{DCR13}) \\ E_H = \max(0, E_H^-, E_H^+) + 2(a-1) \log n.$$

The final term retains the exact diagonal change from coefficients of $t(z)$ to the original remainder in powers of y : $D_n = \text{diag}(1, n, \dots, n^{a-1})$. Equations DCR11–13 prove the high quotient form bounds with center c_{high} and error E_H .

For completeness the low source degrees are $g-1, g$, even though the retained divisor has degree a . Put

$$y_0 = 2Q/\beta, \quad c_{\text{low}}(Q, \beta) = 2Q \log(2Q/\beta) - 2Q, \quad T = 2 \max(1, R_b), \\ C_L^{\text{rem}} = a 2^a (1+R_b)^a \sqrt{g+1} T^g.$$

Here C_L^{rem} denotes a remainder bound, not the Gamma density constant. Cauchy's remainder integral on $|y| = T$ gives $\|\text{rem}_{p_b} f\|_2 \leq C_L^{\text{rem}} \|\text{coef } f\|_2$ for every $\deg f \leq g$: the denominator is at least $(T/2)^a$; the divided-difference coefficient sum is at most $aT^{a-1}(1+R_b)^a$; and $\max |f| \leq$

$\sqrt{g+1}T^g\|\text{coef } f\|_2$. The full low Gram bounds OOQ26–27, whose proofs do not use the divisor, therefore give the error

$$E_L(Q, \beta) = \max \left\{ 0, \log \sum_{j=0}^g \frac{2(2Q+2j)!}{\beta^{2Q+2j+1}} - c_{\text{low}}(Q, \beta), \right. \\ \left. \beta + 2\log(g+1) + \log(2g+1) + g\log 64 \right. \\ \left. + 2g\log(1+y_0) + 2\log C_L^{\text{rem}} \right\}. \quad (\text{DCR14})$$

The lower bound follows by taking the minimum of the full Gram lower bound on the exact remainder fibre; the upper bound uses the canonical remainder lift of degree below a . This proves both low estimates without dropping the b extra relation columns.

5 Return to the unchanged Gamma metric and observation

The full source comparison OOQ31 for order $s = 1$ applies to every polynomial just used, since its degree is at most $q + g \leq 2q$. Its common mass factor is exactly $\beta_1 = (2\pi)^{1/2}/(\sqrt{2}\Gamma(1/2)) = 1$. Keep the OOQ30 positive factors $L_{1/q}, U_{1/q}$ and the complete OSP24 polynomial allowance $\mathcal{E}_k$. Define the same original scalar centers

$$c_{\text{lo}} = 2q\log(2q/\vartheta) - 2q, \quad c_{\text{hi}} = 2q\log(q/2) - qW(2, \vartheta), \quad \Delta c = c_{\text{lo}} - c_{\text{hi}} = \frac{q\mathfrak{C}_\partial}{2}. \quad (\text{DCR15})$$

This last identity is OOQ36 with its unchanged physical source scale. An explicit common nonnegative error is

$$\varepsilon_k^b = \mathcal{E}_k + \max(|\log L_{1/q}|, |\log U_{1/q}|) \\ + \max_{\beta=\beta_\pm} \{E_L(Q, \beta) + E_H(Q, \beta) + |c_{\text{low}}(Q, \beta) - c_{\text{lo}}| + |c_{\text{high}}(Q, \beta) - c_{\text{hi}}|\}. \quad (\text{DCR16})$$

The notation $E_H(Q, \beta)$ means DCR13 with the corresponding equilibrium. Taking the same attained minima in the source comparison now proves

$$e^{c_{\text{lo}} - \varepsilon_k^b} I_a \preceq Q_{b,M} \preceq e^{c_{\text{lo}} + \varepsilon_k^b} I_a \quad (M = g - 1, g), \\ e^{c_{\text{hi}} - \varepsilon_k^b} I_a \preceq Q_{b,M} \preceq e^{c_{\text{hi}} + \varepsilon_k^b} I_a \quad (M = q + g - 1, q + g). \quad (\text{DCR17})$$

All identities are in the original y remainder coordinates. In particular

$$\varepsilon_k^b = O_A(q^{7/8} + k\log(q+2)) = o(q). \quad (\text{DCR18})$$

Indeed $n\rho^2 = n^{7/8}$, $\log C_b = O(g\log q)$, $\log C_L^{\text{rem}} = O(g\log q)$, and the factorial bounds in OOQ28 give $E_L = O(g\log q)$. The mean-value bounds OOQ5–6 give scalar discrepancies $O(k\log q)$, since $Q = q - O(k)$. OOQ30–35 bounds the remaining full-source allowances by this order.

Let $l_i = q - 1 + i$, $u_i = 2q - 1 + i$, $i = 0, 1$. Inverting DCR17 and applying the identical congruence A_b in DCR8 gives

$$e^{\Delta c - 2\varepsilon_k^b} B_0 C_{R,l_i}^d B_0^* \preceq B_0 C_{R,u_i}^d B_0^* \preceq e^{\Delta c + 2\varepsilon_k^b} B_0 C_{R,l_i}^d B_0^*. \quad (\text{DCR19})$$

For example both lower and upper covariances are bounded by the displayed scalars times $A_b A_b^*$, which proves the comparison. The exact possibly ill-conditioned map A_b therefore cancels between endpoints. No bound on its condition number was assumed.

6 The distant primary components are paid for in the full metric

Use the same reference coefficient form R_Z as CGR2 and the proved global constants

$$l_d R_Z \preceq C_{R,N}^d \preceq u_{d,N} R_Z$$

of CGR17 at all four endpoints. Those constants are explicit there, retain every root, and satisfy $\log^+(1/l_d) + \log^+ u_{d,N} = O_A(q \log(q+2))$. Take a basis orthonormal for this reference with its first $r-h$ rows spanning DCR6. The remaining Schur complement has dimension h . Its metric lies between $l_d I_h$ and $u_{d,N} I_h$, because these bounds pass to the attained minimum on every fibre. Enlarging the source adds positive covariance terms, so its high/low Schur determinant ratio lies between 1 and $(u_{d,u_i}/l_d)^h$. Define

$$E_i^d = 2(r-h)\varepsilon_k^b + h\{\log^+(u_{d,u_i}/l_d) + |\Delta c|\}.$$

The determinant identity and DCR19 prove the full return

$$\left| \log \frac{\det C_{R,u_i}^d}{\det C_{R,l_i}^d} - r\Delta c \right| \leq E_i^d = o_A(kq). \quad (\text{DCR20})$$

The empty first block has determinant one if $r-h=0$. For the order bound use $h \leq b_{\max} = O_A(q^{1/8})$, $r = O(k)$, and DCR18. A finite rate bound is $O_A(kq^{7/8} + q^{9/8} \log(q+2) + q \log(q+2))$. Thus the distant roots have not been discarded at zero cost. Adding the two evaluated returns yields

$$\left| \sum_{i=0}^1 \log \frac{\det C_{R,u_i}^d}{\det C_{R,l_i}^d} - rq\mathfrak{C}_\partial \right| \leq E_0^d + E_1^d = o_A(kq). \quad (\text{DCR21})$$

7 The actual graph correction now has one four-angle term

Retain the original matrices of CGS13–19,

$$W_N = B_F A^{-1}, \quad \tilde{C}_N = W_N W_N^*, \quad T_N = W_N^* \tilde{C}_N^{-1/2}, \quad S_N = T_N^* P_{I_N} T_N, \quad I_N = \chi' \mathcal{P}'_{N-v-q'}.$$

The full conductor inverse, actual row space and lower-source projection are all unchanged. CGS20 states

$$|\log(\det C_{R,N}^U / \det R_Z) - \log \det S_N| \leq K_N = O_A(q \log(q+2)).$$

CG24 gives $C_{R,l_i}^d = C_{R,l_i}^U$ at both low cutoffs, and CG32 supplies the complete finite graph error B_{CG32} . Substitution of DCR20 into that exact identity proves

$$\left| \mathcal{G}_K - rq\mathfrak{C}_\partial - \log \frac{\det S_{q-1} \det S_q}{\det S_{2q-1} \det S_{2q}} \right| \leq B_{\text{CG32}} + E_0^d + E_1^d + \sum_{N=q-1, q, 2q-1, 2q} K_N = o_A(kq). \quad (\text{DCR22})$$

In fact CG32 has sign $\mathcal{G}_K = \sum_i (\log \det C_{R,u_i}^d - \log \det C_{R,u_i}^U)$ up to its stated error. Replace the first logarithm by its low partner plus $r\Delta c$, use the two exact low equalities, then CGS20 at all four endpoints. This proves every sign in DCR22. The value $r = 8k - 32 - v + s_k$ remains exact.

The outer covariance part is now evaluated at the required scale. The four matrices S_N are the remaining original projection-angle calculation. Their determinants are positive, but this proof does not assign the sign or leading coefficient of their four-return. Nor does a positive covariance result evaluate a complex projected current.

8 The receiver in the original arithmetic kernel

The same proof evaluates the entire g -dimensional outer metric. Here are the needed full-metric constants; the previous restricted covariance bounds are not assumed to apply to the identity observation. In the original S' coefficient frame put

$$\widehat{C}_N = (Q_N^d)^{-1}, \quad \widehat{l} = (h_+^0)^{-1}, \quad \widehat{u}_N = B_{\text{rem}}(M)^2/h_-(M), \quad (\text{DCR23})$$

where the complete explicit $B_{\text{rem}}(M), h_-(M), h_+^0$ are CGR16 with $M = N - q + g$. Then $\widehat{l}I_g \preceq \widehat{C}_N \preceq \widehat{u}_N I_g$. For the lower bound, at $N = q - 1$ the full quotient is the whole degree-below- g source. Its Gram has norm at most h_+^0 ; taking the inverse gives the lower bound. Covariance increases under source enlargement, so it holds at all four cutoffs. For the upper bound the literal remainder covariance is $R_M H_M^{-1} R_M^*$; CGR16 proves $\|R_M\| \leq B_{\text{rem}}(M)$ and $H_M \succeq h_-(M)I$. This proves the upper bound. Its logarithmic costs are $O_A(q \log(q+2))$, by the explicit factors of CGR16. Thus every full-metric constant has been supplied before deleting a primary component.

Apply DCR5–19 with the identity observation on E_d . Its good row subspace is the full dual of E_b , has dimension $g - b$, and its complementary Schur rank is exactly b . Use the same physical transport DCR1. Define

$$\widehat{E}_i = 2(g - b)\varepsilon_k^b + b\{\log^+(\widehat{u}_{u_i}/\widehat{l}) + |\Delta c|\}.$$

The complete DCR20 argument with the now-proved DCR23 bounds gives

$$\left| \log \frac{\det Q_{l_i}^d}{\det Q_{u_i}^d} - g\Delta c \right| \leq \widehat{E}_i, \quad \left| \mathcal{R} \log \det Q_N^d - gq\mathfrak{C}_\partial \right| \leq \widehat{E}_0 + \widehat{E}_1 = o_A(kq). \quad (\text{DCR24})$$

The reciprocal determinant accounts for the sign in the metric return.

Let J_K be any fixed low-polynomial frame for $\overline{K} \subset \mathbb{C}^g$, as in KF45 and CG30, and choose a fixed right inverse J_R of the actual A_R . The square frame $T = [J_K, J_R]$ is invertible because $\ker A_R = \overline{K}$ and $A_R J_R = I_r$. Schur complementation in $T^* Q_N^U T$ gives exactly

$$\det(J_K^* Q_N^U J_K) = |\det T|^2 \det Q_N^U \det C_{R,N}^U. \quad (\text{DCR25})$$

Indeed the complementary Schur metric is $(A_R(Q_N^U)^{-1} A_R^*)^{-1}$; its constrained minimum is proved by the same orthogonal-lift calculation as DCR8. This verifies the identity for every actual frame, not only for an orthonormal basis. The fixed $|\det T|^2$ cancels with the four original signs.

Write E_N^{vol} for the finite CG24 bound

$$E_N^{\text{vol}} = 2v(L+1) \log \frac{q+L}{q-v+1} + B_N \quad (L = N - q = q - 1, q),$$

and put $E_N^{\text{vol}} = 0$ at the two low cutoffs, where CG24 proves exact equality $Q_N^U = Q_N^d$. Let B_{KF49} be the entire displayed finite right side of KF49, including $2mL_k^{\text{form}}$, the actual low-degree angle bound and all source-map exterior allowances. Then the *original arithmetic* kernel metric obeys

$$\left| \mathcal{R} \log \det H_{K,N}^{\text{ar}} - gq\mathfrak{C}_\partial - \log \frac{\det S_{q-1} \det S_q}{\det S_{2q-1} \det S_{2q}} \right| \leq B_{\text{KF49}} + \widehat{E}_0 + \widehat{E}_1 + \sum_N (E_N^{\text{vol}} + K_N) = o_{A,h}(kq). \quad (\text{DCR26})$$

For the proof first apply KF49 to its exact original arithmetic kernel frame $I_K = V^{-1}\pi^T$, with $Z = [M_A, S_k Z_k]$. Apply DCR25 to the transported actual plane, use CG24 on its full metric determinant, use DCR24, and finally apply CGS20 to the covariance term at each cutoff. Every

error displayed above is the error of one of these exact maps, so the triangle inequality proves DCR26. The same divisor h , period and physical units remain in B_{KF49} .

In particular the independent covariance calculation has now been carried all the way back to the object in MR1–23. The coefficient multiplying the evaluated scalar is the exact $g = 16k - 48 - v$. It has not been replaced by $m = 8k - 16$. The difference is retained by the *actual* four-angle term; its leading value is still required before the requested allocation can be assigned.

One evaluated consequence uses no unknown angle coefficient. For each i , source enlargement makes $C_{R,u_i}^U \succeq C_{R,l_i}^U$, since the observation functional annihilates every complete relation (CGR6). Consequently CGS20 gives

$$\log \frac{\det S_{l_i}}{\det S_{u_i}} \leq K_{l_i} + K_{u_i}.$$

The original kernel metric itself decreases with source enlargement. Therefore, with $\mathcal{B}_k$ the explicit right side of DCR26,

$$0 \leq \mathcal{R} \log \det H_{K,N}^{\text{ar}} \leq gq\mathfrak{C}_{\partial} + \mathcal{B}_k + \sum_N K_N. \quad (\text{DCR27})$$

The lower bound follows by pairing the identical restricted metric at each low/high pair before taking its determinant. This bounds the actual kernel from above at the requested scale while retaining the leading deficit as the four-angle determinant. It supplies neither its missing value nor the individual complex current signs.

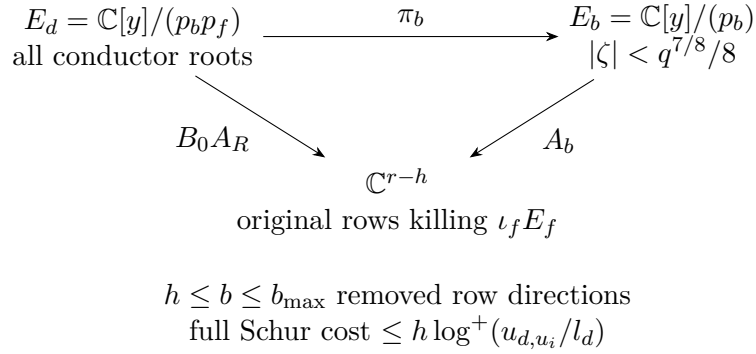


Figure 2: The exact factorization DCR5–8. The diagram uses the original quotient and fixed observation. DCR10–19 evaluates the retained return, and DCR20 pays for every complementary row. The division circle defines primary factors with full multiplicity; it is not a replacement of the physical root coordinates. The complete final receiver is DCR22.

References

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